

Yash Patel

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EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Applied Science in Mechatronics Engineering

Sept. 2022 – May 2027

Relevant Courses: Digital Computation, Computer Structures and RTOS, Data Structures and Algorithms, Linear Systems and Signals, Microprocessors and Digital Logic, Microprocessor Systems and Interfacing

EXPERIENCE

Autonomous Robotics SWE Intern

Sept. 2025 – Dec. 2025

Daylun Housing

Toronto, ON

- Designed and implemented a **path-planning algorithm** for an autonomous mobile robot (AMR) to navigate multiple assembly stations for modular housing panel construction.
- Developed an **autonomous pick-and-place system** using **ROS2**, **Gazebo**, and **MoveIt**, including writing custom robot controllers from scratch to handle stud placement on the AMR.
- Engineered **autonomous nailing**, **spray-foam**, and **steel-strapping systems** to automate panel processes.
- Built and simulated the **entire AMR workflow in NVIDIA IsaacSim**, coding each station's logic and motion sequence for end-to-end system validation.

CAD/CAM Programmer

Jan. 2025 – Apr. 2025

Sable Metal Manufacturing

Markham, ON

- Programmed sheet metal **bending sequences** using **INFORM II** for the **Motoman HP-50** robotic arm to automate production, resulting in a 15% increase of efficiency.
- Designed 3D models of sheet metal components for **laser cutting**, **bending**, and **welding** using **SolidWorks**.
- Structured robotic programs into three distinct phases: **picking**, **bending**, and **stacking**, to optimize process clarity and execution.

Mechanical/Manufacturing Engineering Intern

Sep. 2023 – Dec. 2023

Linamar Corporation

Guelph, ON

- Automated part-sorting workflows using Excel VBA to reduce processing time for monthly deadlines.
- Designed optimized floor layouts for new plant areas using **AutoCAD**, reducing part retrieval time by **12%**.

PROJECTS

2024 Not-A-Boring Competition Navigation Challenge

Oct. 2024 – Apr. 2024

- Developed a **React**-based GUI to subscribe to **ROS2** nodes and display real-time data from published sources, enhancing the user interface for autonomous vehicle monitoring.
- Implemented a **SLAM-based localization system** using **Python** and **Ultra WideBand (UWB)** technology to accurately track the robot's position relative to UWB anchor points.
- Created **mounts** and **housings** for Ultra Wideband sensors, batteries, and motor drivers using **SolidWorks**.
- Designed the vehicle chassis using **SolidWorks** along with a **milling machine** and **lathe**.
- Achieved **1st** place at the competition, was held in Austin, Texas.

2025 Not-A-Boring Competition

May 2024 – Apr. 2025

- Engineered a micro-scale **Tunnel Boring Machine (TBM)** for the **2025 Not-A-Boring Competition**.
- Developed a **control system** to regulate the excavation rate of the TBM by adjusting the powertrain motor through a **Variable Frequency Drive (VFD)**, ensuring optimal propulsion and excavation efficiency.
- Designed the **motor**, **gearbox**, and **couplings** in **Onshape** to withstand the 2200 Nm torque requirement.
- Utilized **rapid prototyping** to create enclosures and mounts for above and below ground PCB control stations.

Custom Programming Language

Aug. 2024

- Developed a **python** interpreter to execute **stack-based programs**, supporting primarily arithmetic operations.
- Created a **compiler** that parses **opcodes** (PUSH, POP, etc.) and generates **p-code** for program execution.

SKILLS

Programming Languages: C++, Python, RobotC, Java, Excel VBA, MATLAB, SQL, Ladder Logic, INFORM II

Web Development & Libraries: HTML, CSS, JavaScript, React, OpenCV, Tensorflow, Pytorch

Tools, Frameworks, & Technologies: PLC, YOLOv8, JIRA, Git, Linux, ROS2, Verilog, Docker, MS Office, STM32